Sentiment Classification of Kinyarwanda Tweets: Comparing Classical, Recurrent and Transformer Models on AfriSenti

YOUR FULL NAME · Machine Learning / NLP - Summative Project

GitHub Repository	https://github.com/YOUR_GITHUB_USERNAME/kin-sentiment
Demo Video	https://YOUR-VIDEO-LINK
Live System	https://huggingface.co/spaces/YOUR_HF_USERNAME/kin-sentiment

Abstract

Sentiment analysis tools rarely exist for African languages, even widely spoken ones such as Kinyarwanda. This project builds and deploys a three-way (positive / neutral / negative) sentiment classifier for Kinyarwanda tweets using the Kinyarwanda portion of the AfriSenti benchmark (Muhammad et al., 2023a), which contains 5,155 human-annotated tweets. A TF-IDF word-unigram logistic regression baseline achieved 57.9 test macro-F1. Adding character n-grams, which capture the agglutinative morphology of Kinyarwanda, raised this to 62.3 (E6, Linear SVM), a statistically significant improvement (McNemar p = 0.0002). BiLSTM models trained from scratch performed worse than the baseline (57.2), which we attribute to the 31.2% out-of-vocabulary rate and small training set. Fine-tuning of pretrained African-centric Transformers (AfriBERTa, AfroXLMR) is provided as a reproducible GPU notebook. Error analysis shows that most mistakes involve the neutral class and that human annotators themselves agreed unanimously on only 35.8% of tweets (Fleiss' κ = 0.318), suggesting label ambiguity limits achievable accuracy. The final model is deployed as a Gradio web application.

1. Introduction

Social media is now a primary channel through which people in Rwanda discuss public services, news, sport, entertainment and politics. Organisations such as government agencies, telecom companies, banks and media houses increasingly want to understand public opinion expressed online, but almost all off-the-shelf sentiment tools are built for English and a few other high-resource languages. Joshi et al. (2020) show that the vast majority of the world's languages, including most African languages, are severely under-represented in NLP resources and research. Kinyarwanda, the national language of Rwanda, is one of these: despite a large speaker population, labelled datasets are small and only a few pretrained models include it (Niyongabo et al., 2020; Nzeyimana & Niyongabo Rubungo, 2022).

Kinyarwanda is a Bantu language with rich agglutinative morphology: a single word can combine subject, tense, object and negation markers around a stem (Nzeyimana & Niyongabo Rubungo, 2022). For example, the gratitude expressions *urakoze*, *murakoze* and *mwakoze* ("thank you", singular/plural/past) are different surface words sharing the stem -koze. This makes word-level models sparse and motivates sub-word and character-level representations. Tweets add further challenges: informal spelling, elongated characters, emojis and frequent code-switching with English and French.

Objective. The goal of this project is to build, evaluate and deploy a sentiment classifier that assigns Kinyarwanda text to one of three classes (positive, neutral, negative), and to understand which modelling choices matter for a low-resource, morphologically rich language. Specifically, we ask: (RQ1) Does a meaningful model beat a simple bag-of-words baseline? (RQ2) Do character-level features help, given Kinyarwanda morphology? (RQ3) Do neural sequence models trained from scratch help on only ~3,300 training tweets? (RQ4) Does African-centric pretraining help compared with generic multilingual

pretraining? (RQ5) Which kinds of tweets remain hard, and why? **Target users** are Rwandan organisations and researchers who want to monitor public sentiment in Kinyarwanda, e.g. customer feedback for a telecom or public reaction to a government announcement.

2. Related Work

African-language sentiment analysis. AfriSenti (Muhammad et al., 2023a) is the largest sentiment benchmark for African languages, with over 110,000 tweets in 14 languages annotated by native speakers, and was the basis of the SemEval-2023 Task 12 shared task (Muhammad et al., 2023b). In the shared task, the strongest Kinyarwanda systems used pretrained Transformers: the organisers' AfroXLMR-large baseline reached 70.6 weighted-F1 and the best Kinyarwanda team result (72.5 weighted-F1) used KinyaBERT, a morphology-aware Kinyarwanda language model (Muhammad et al., 2023b; Nzeyimana & Niyongabo Rubungo, 2022). For Kinyarwanda text classification more broadly, Niyongabo et al. (2020) released KINNEWS, a 14-class news corpus, and showed that classical models with TF-IDF features are competitive with neural models when data is limited.

Classical vs neural text classification. Linear models over n-gram features remain strong baselines for short-text sentiment; Wang and Manning (2012) showed that Naive Bayes and SVM variants with bigrams are hard to beat on short snippets, and Joulin et al. (2017) showed that simple linear models with sub-word n-grams rival deep networks at a fraction of the cost. Recurrent networks such as LSTMs (Hochreiter & Schmidhuber, 1997) with attention (Bahdanau et al., 2015) model word order, but typically need more data to learn good embeddings from scratch.

Pretrained multilingual Transformers. mBERT (Devlin et al., 2019) and XLM-R (Conneau et al., 2020) are pretrained on 100+ languages but include little or no Kinyarwanda. African-centric models address this: AfriBERTa (Ogueji et al., 2021) was pretrained from scratch on under 1 GB of text in 11 African languages including Kinyarwanda/Kirundi, and AfroXLMR (Alabi et al., 2022) adapts XLM-R to 17 African languages through multilingual adaptive fine-tuning (MAFT).

Gap addressed. Most published Kinyarwanda sentiment systems are shared-task submissions that report a single leaderboard score. This project contributes a controlled comparison of classical, recurrent and Transformer approaches under the same splits and preprocessing, with ablations, significance testing and an analysis of annotator disagreement, and it packages the result as a deployed, usable application.

3. Dataset and Data Preparation

Source and licence. We use the Kinyarwanda subset of AfriSenti (Muhammad et al., 2023a), downloaded from the official AfriSenti-SemEval GitHub repository (CC BY 4.0). The tweets are real, naturally occurring posts collected from Twitter and labelled by three native-speaker annotators; the final label is the majority vote. No synthetic or machine-generated data is used. We keep the official train / dev / test split so that results are comparable with the literature.

Table 1. AfriSenti-Kinyarwanda splits and label distribution.

Split	Tweets	Negative	Neutral	Positive
Train	3302	1146	1257	899
Dev (validation)	827	287	315	225
Test	1026	355	393	278
Total	5155	1788	1965	1402

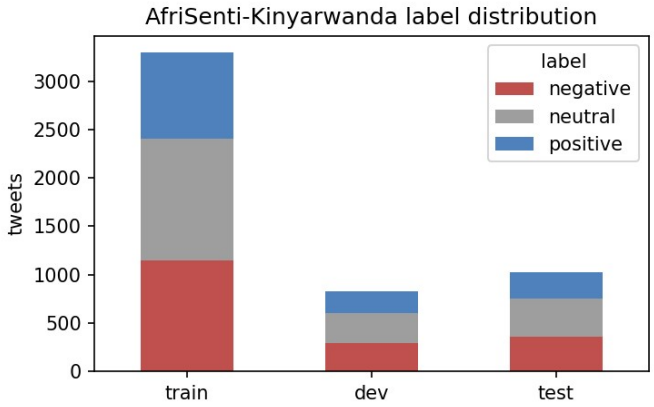


Figure 1. Label distribution per split. Classes are moderately imbalanced (positive is smallest).

Characteristics and quality issues. Inspecting the data revealed: (i) many replies begin with long chains of anonymised mentions (up to 11 "@user" tokens); (ii) URLs, hashtags and elongated words such as *Hhhhhh* or *Yoooo*; (iii) emojis in 29.6% of test tweets; (iv) code-switching with English/French in about 11.4% of test tweets; (v) many news-style and promotional tweets whose sentiment is implicit. The data files are **sorted by label**, so shuffling during training is essential. Most importantly, the individual annotations released with AfriSenti show that the three annotators gave the same label to only 35.8% of Kinyarwanda tweets, with Fleiss' $\kappa = 0.318$ ("fair" agreement; Fleiss, 1971). Annotators also used 'mixed' (1567 labels) and 'indeterminate' (466) labels that were later collapsed. Label noise is therefore substantial and sets a ceiling on achievable performance.

Preprocessing (src/preprocess.py). For classical and BiLSTM models: lower-casing; URLs → *xxurl*; runs of @-mentions → a single *xxuser* token (keeping the "this is a reply" signal); any character repeated 3+ times is capped at two (*hhhhhh* → *hh*); emojis are kept and separated into their own tokens; remaining punctuation is removed. We deliberately **do not remove stopwords**, because the standard Kinyarwanda stopword list (Niyongabo et al., 2020) contains the negation word *nta*, and removing negation can flip sentiment; this decision is tested in ablation A3. For Transformers we apply only light cleaning (mentions, URLs, elongation), leaving case, punctuation and emojis to the model's sub-word tokenizer.

4. Methodology

All models map a tweet to one of three labels. Hyper-parameters are selected only on the dev set; the test set is used once per configuration for reporting. Because the classes are imbalanced and all three matter equally, **macro-F1** is our primary metric; we also report accuracy and weighted-F1 (the official AfriSenti metric) for comparability.

4.1 Baseline

E1: TF-IDF word unigrams + logistic regression. Each tweet is represented as a sparse vector of sub-linear TF-IDF weights over its words, and a multinomial logistic regression with balanced class weights is trained on top (regularisation strength $C \in \{0.1, 0.3, 1, 3, 10\}$ tuned on dev). This is the standard, transparent starting point for text classification and was also used as a baseline for KINNEWS (Niyongabo et al., 2020). A majority-class predictor (E0) is reported as a floor.

4.2 Classical experiments

Building on E1 we vary one factor at a time: word bigrams (E2); **character n-grams** of length 2–5 within word boundaries (E3), which let the model share information across morphological variants such as

urakoze/murakoze/mwakoze; the union of word 1–2-grams and character 2–5-grams (E4); and the classifier on the E4 features: Multinomial Naive Bayes (E5) and a linear SVM (E6). Ablations test preprocessing choices on the E4 configuration: raw text (A1), emojis removed (A2), and stopwords removed (A3).

4.3 BiLSTM trained from scratch

To test whether modelling word order helps, we implemented a BiLSTM classifier in PyTorch (src/bilstm.py). Input words are mapped to indices (vocabulary built from training words with frequency ≥ 2; 4,345 types; unknown words → <unk>), passed through a 100-dimensional embedding layer learned from scratch, dropout (0.5), and a bidirectional LSTM with 64 hidden units per direction. The 128-dimensional hidden states are pooled into one sentence vector either by **max-pooling** over time (N1) or by **additive attention** (N2), where each state h_t receives a weight $a_t = \text{softmax}(v^T \tanh(W h_t))$ and the sentence vector is $\Sigma\, a_t h_t$ (Bahdanau et al., 2015). A linear layer maps it to three logits. Training uses cross-entropy loss, Adam (learning rate 2×10^{-3}), batch size 32, gradient clipping at 1.0, and early stopping on dev macro-F1 (patience 3, max 30 epochs). N3 doubles the embedding (200) and hidden (128) sizes. Each configuration is run with three random seeds and we report mean ± standard deviation.

4.4 Fine-tuned Transformers

We fine-tune three pretrained encoders with the Hugging Face Transformers library (Wolf et al., 2020) (src/finetune_transformer.py): **T1 mBERT** (Devlin et al., 2019), whose pretraining data does not include Kinyarwanda; **T2 AfriBERTa-base** (Ogueji et al., 2021); and **T3 AfroXLMR-base** (Alabi et al., 2022). The tweet is split into sub-word tokens (max 128), the final-layer representation of the first token is fed to a randomly initialised linear classification head, and all weights are updated (full fine-tuning) with AdamW, weight decay 0.01, linear warm-up over 10% of steps, 5 epochs, batch size 32, learning rate 3×10^{-5} (5×10^{-5} for AfriBERTa), mixed precision on a Colab T4 GPU. The checkpoint with the best dev macro-F1 is kept, and each model is run with three seeds. Sub-word tokenization addresses the out-of-vocabulary problem that limits the word-level BiLSTM.

4.5 Own work vs existing resources

Table 2. Distinction between existing resources and own implementation.

Existing resources used	Implemented in this project
AfriSenti-Kin data and annotator labels (Muhammad et al., 2023a)	Data inspection, cleaning pipeline, agreement analysis
Kinyarwanda stopword list (Niyongabo et al., 2020)	Baseline, 10 classical experiments and ablations with dev-set tuning
scikit-learn, PyTorch, Hugging Face Transformers libraries	BiLSTM + attention model and training loop written from scratch
Pretrained mBERT, AfriBERTa, AfroXLMR checkpoints	Fine-tuning script, multi-seed evaluation, model selection
Gradio, Hugging Face Spaces	Error analysis, significance tests, figures, web app, this report

5. Experiments and Results

Table 3. All experiments (scores ×100). M-F1 = macro-F1, W-F1 = weighted-F1. Neural rows: mean ± std over 3 seeds.

ID	Approach	Best param	Dev M-F1	Test Acc	Test M-F1	Test W-F1
E0	Majority class (neutral)	-	18.4	38.3	18.5	21.2
E1	**TF-IDF word unigrams + LogReg**	**1.0**	**59.8**	**57.4**	**57.9**	**57.4**
E2	TF-IDF word 1-2grams + LogReg	10.0	58.6	58.2	58.7	58.3
E3	TF-IDF char 2-5grams + LogReg	1.0	63.2	61.1	61.3	60.9
E4	Word 1-2 + char 2-5 + LogReg	10.0	64.4	62.0	62.4	61.9
E5	Word 1-2 + char 2-5 + Multinomial NB	0.3	64.2	61.6	62.0	61.7
E6	**Word 1-2 + char 2-5 + Linear SVM**	**0.1**	**64.7**	**62.0**	**62.3**	**61.8**
A1	E4 on RAW text (no preprocessing)	1.0	63.9	61.1	61.5	61.0
A2	E4 with emojis REMOVED	1.0	63.6	61.2	61.7	61.1
A3	E2 with stopwords REMOVED (word only)	0.3	58.0	55.9	56.5	55.7
N1	BiLSTM + max-pooling (emb 100, hid 64)	3 seeds	58.4	57.3	57.2 ± 1.5	57.2
N2	BiLSTM + additive attention (emb 100, hid 64)	3 seeds	58.3	55.8	56.1 ± 0.7	55.8
N3	BiLSTM + attention, bigger (emb 200, hid 128)	3 seeds	57.7	55.3	55.3 ± 0.2	55.1

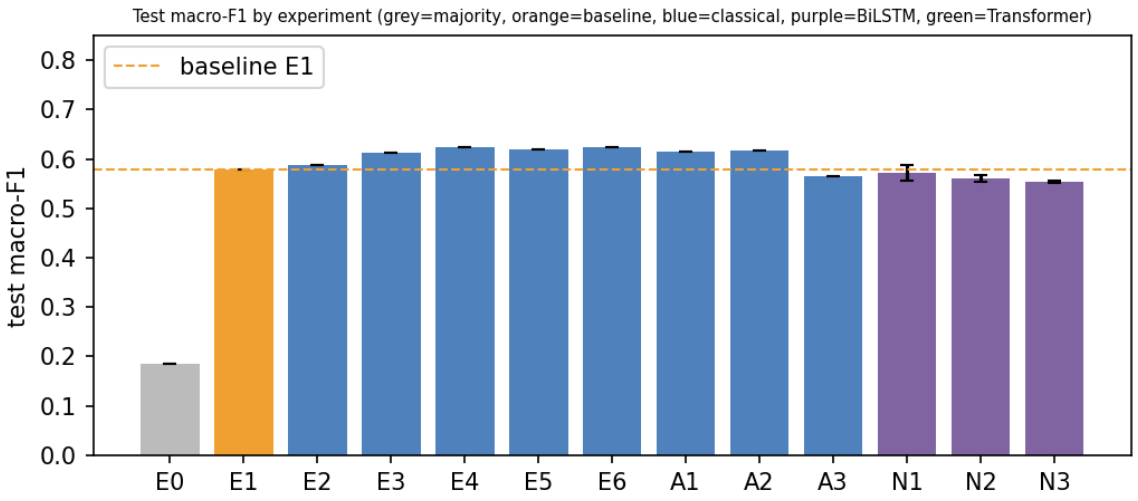


Figure 2. Test macro-F1 per experiment; dashed line = baseline E1. Error bars = std over seeds.

RQ1 – baseline vs better models. The majority-class floor (E0) scores only 18.5 macro-F1, while the word-unigram baseline reaches 57.9. The best classical model, E6, improves this by 4.4 points to 62.3 (95% bootstrap CI 59.4–65.3 vs 54.8–61.0 for the baseline). On the paired test predictions it fixes 103 baseline errors while breaking 56, and McNemar's exact test gives p = 0.0002, so the improvement is unlikely to be chance.

RQ2 – character n-grams. Word bigrams alone barely help (E2: 58.7), because with only 3,302 tweets most bigrams occur once. Switching to character 2–5-grams gives the largest single jump (E3: 61.3, +3.4 points). This supports our hypothesis: character n-grams recover shared stems and affixes of agglutinative Kinyarwanda words (e.g. -koze in all 'thank you' variants) and are robust to spelling variation, so a word never seen in training can still activate useful features. Combining word and character features (E4) adds a further gain (62.4). With these features the choice of classifier matters little: Naive Bayes, logistic regression and SVM are within about 0.4 points, i.e. representation matters more than the classifier.

Ablations. Keeping emojis helps: removing them (A2) lowers dev macro-F1 from 64.4 to 63.6, and in the error analysis tweets containing emojis are classified more accurately (65.8% vs 60.4%). Our cleaning gives a small gain over raw text (A1: 63.9 dev), mainly by collapsing mention chains and elongations. Removing stopwords hurts (A3: 56.5 vs 58.7 test for the same model), consistent with our concern that the list removes the negator *nta*. Note that with 827 dev tweets, differences below ~1–2 points are within noise.

RQ3 – BiLSTM from scratch. All BiLSTM variants performed **worse than the linear baseline** (best N1: 57.2 ± 1.5), and attention (N2) or a larger network (N3) did not help. We investigated why: 31.2% of test tokens are out of the training vocabulary, so a third of each tweet reaches the LSTM as the same <unk> vector, whereas character n-grams still extract information from those words. With only ~3,300 short tweets, the embedding layer (≈430k parameters) cannot learn good word representations, and the dev score peaks within 6–13 epochs before over-fitting. The larger N3 over-fit fastest (best epoch 6–7). This matches the observation of Niyongabo et al. (2020) that classical models are competitive on small Kinyarwanda datasets, and shows that model complexity is not a substitute for data.

RQ4 – pretrained Transformers. [Run notebooks/finetune_transformers_colab.ipynb, copy results/transformer_*.csv into results/ and rebuild this report to fill in this paragraph and the T1–T3 rows of Table 3.] For reference, the AfriSenti AfroXLMR-large baseline reported 70.6 weighted-F1 on this test set (Muhammad et al., 2023b).

6. Error Analysis and Discussion

Table 4. Per-class test results for E6.

Class	Precision	Recall	F1
negative	61.1	67.6	64.2
neutral	60.1	53.9	56.8
positive	65.7	66.2	65.9

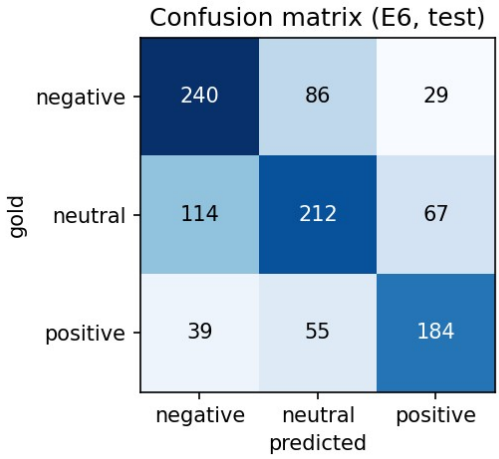


Figure 3. Confusion matrix of E6 on the test set.

Where the model fails. E6 makes 390 errors on 1,026 test tweets. Neutral is the hardest class (F1 56.8) and the two most frequent confusions both involve it: neutral→negative (114) and negative→neutral (86). Direct polarity flips (positive ↔ negative) are rare (68), so the model has learned polarity but struggles with its *intensity* and with implicit sentiment. We grouped test tweets by properties (Table 5) to find systematic weaknesses.

Table 5. E6 accuracy on subsets of the test set (OOV = share of words unseen in training).

Tweet property	n	Accuracy	Accuracy (others)
≤ 5 content words (short)	74	56.8%	62.4%
Contains a question mark	180	55.6%	63.4%
Code-switched (English/French words)	117	59.8%	62.3%
Contains emoji	304	65.8%	60.4%
OOV rate ≤ 10%	162	66.0%	-
OOV rate 10–30%	516	63.9%	-
OOV rate > 30%	348	57.2%	-

Accuracy falls steadily as the share of unseen words rises (from 66.0% to 57.2%), confirming that vocabulary coverage is the main bottleneck for a 3,300-tweet training set. Questions, which often express sarcasm or rhetorical complaint without explicit sentiment words, and very short tweets, which give few features, are also harder. Code-switched tweets are slightly harder because English/French words are sparse in the training data.

Representative examples (English glosses are approximate):

- **Correct, high confidence:** *"Urakoze cyane nawe umunsi mwiza 🥰"* ("Thank you very much, have a good day too") → positive. Formulaic gratitude (*urakoze/murakoze*), *umunsi mwiza* ("good day") and positive emojis are strong, frequent cues.
- **Negation (bag-of-n-grams limitation):** *"Ntacyo itwaye ikibazo ndumva ari battery😅"* ("It doesn't matter, I think the problem is the battery") – gold neutral, predicted negative. *ikibazo* ("problem") is a strong negative feature, and a linear model over n-grams cannot compose it with *ntacyo itwaye* ("it's nothing").
- **Implicit sentiment needing world knowledge:** *"Inama y'Abaminisitiri yatangaje ko bitakiri itegeko kwambara agapfukamunwa"* ("The Cabinet announced that wearing masks is no longer mandatory") – gold positive, predicted neutral. The tweet is written in neutral news style; it is positive only because readers know the news is welcome.
- **Negative news in neutral style:** *"Rutahizamu … yahagaritswe iminsi 8 … kubera imyitwarire itari myiza"* ("The striker has been suspended for 8 days for bad behaviour") – gold negative, predicted neutral. Similar sports/news reports are labelled neutral elsewhere in the data, which illustrates annotator inconsistency.
- **Humour and emojis:** *"…none banz nkicyonka😭😭😭"* – gold neutral, predicted negative. Crying emojis are used humorously here, but the model has learned 😭 as a negative cue.
- **Religious text:** *"Wiringire Uwiteka n'umutima wawe wose…"* (Proverbs 3:5, "Trust in the Lord with all your heart") – gold neutral, predicted positive. Religious vocabulary is positive in tone, but annotators judged the verse as neutral.

Relation to label quality and the literature. Many 'errors' are tweets on which humans also disagree. Only 35.8% of Kinyarwanda tweets received three identical labels (κ = 0.318), and the neutral/negative boundary in news-style tweets is exactly where annotators diverge. Thus the achievable score is well below 100%, which helps explain why even the best SemEval systems plateau around 70–72 weighted-F1 on Kinyarwanda (Muhammad et al., 2023b). The remaining gap between our classical models and those systems is consistent with the value of pretrained, morphology-aware representations (Nzeyimana & Niyongabo Rubungo, 2022; Alabi et al., 2022), which can supply knowledge of words never seen in our small training set.

7. Deployment

The system is deployed as a Gradio (Abid et al., 2019) web application hosted on Hugging Face Spaces: https://huggingface.co/spaces/YOUR_HF_USERNAME/kin-sentiment. On start-up the app loads the best classical model (E6), saved as a scikit-learn pipeline (models/best_classical.joblib, ~4 MB). When a user types Kinyarwanda text, the app applies exactly the same preprocessing used in training, runs the model, and displays the predicted label with a score for each class, together with the preprocessed text the model actually saw (useful for explaining errors). Scores come from a softmax over the SVM decision values and are relative confidences rather than calibrated probabilities; a low-confidence warning appears when no class exceeds 0.5. The app can switch to a fine-tuned Transformer from the Hugging Face Hub by setting one environment variable (MODEL_ID). A lightweight linear model was chosen for the default deployment because it runs instantly on the free CPU tier and its behaviour is easy to explain. If the Space is asleep, the app can be run locally with `python app/app.py` (see README).

8. Limitations and Future Work

- **Data size and label noise.** 3,302 training tweets with fair inter-annotator agreement limit every model. Future work: train with soft labels from the three individual annotations, or remove highly disputed tweets from training.
- **Domain.** All data are tweets from a specific period (e.g. COVID-19, CHOGM 2022, sport); performance on customer reviews, SMS or news comments is untested and likely lower.
- **Model limitations.** Bag-of-n-gram models cannot handle negation scope, sarcasm or implicit sentiment. Morphology-aware pretrained models such as KinyaBERT (Nzeyimana & Niyongabo Rubungo, 2022), larger models (AfroXLMR-large) and further pretraining on unlabelled Kinyarwanda tweets are promising.
- **Evaluation.** A single fixed test set of 1,026 tweets gives confidence intervals of about ±3 points; cross-validation and evaluation by additional native speakers on new data would give a more reliable picture.
- **Deployment.** The free hosting tier sleeps when idle, and the app does not yet log feedback from users, which could be used to collect new labelled data.

9. Conclusion

This project built and deployed a Kinyarwanda tweet sentiment classifier. The best model, E6 (TF-IDF word+char n-grams + Linear SVM), reached 62.3 macro-F1 and 61.8 weighted-F1, significantly above a word-level TF-IDF baseline (57.9). The main lessons were: (1) for a morphologically rich, low-resource language, the input representation matters more than the classifier, and character n-grams give most of the gain; (2) a more complex neural model trained from scratch (BiLSTM with attention) did not beat the baseline, because of high out-of-vocabulary rates and little data; (3) seemingly harmless preprocessing such as stopword removal can hurt by deleting negation; and (4) a large share of remaining errors lie on the neutral boundary, where human annotators also disagree. Pretrained, African-centric and morphology-aware language models, more data and better handling of label uncertainty are the clearest routes to improvement.

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